



SPACE DESIGN LOAD SUMMARY

2026-09-27

Ensuite_5

Level: Level 0 · Internal condition: Studio - Ensuite_5

DESIGN CRITERIA

	Heating	Cooling
Room set point	-50.0 °C	150.0 °C
Outdoor dry bulb	—	—
Outdoor RH (coincident)	—	—
ROOM HUMIDITY		
Humidification set point	—	
lower RH limit		
Dehumidification set point	—	
upper RH limit		

SIZING (TAS DESIGN LOADS)

	Heating	Cooling
Design load	0 W	0 W
Design load per area	0.0 W/m ²	0.0 W/m ²
Sizing multiplier	not set	not set
Design loads: from Tas sizing — date/currency not recorded		

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak sensible load	1,090 W	520 W
Peak hour	18:00–19:00	1 Jan 00:00–01:00 (HOY 1)
Room dry bulb	16.3 °C	15.4 °C
Room resultant	14.5 °C	14.3 °C
Room RH	19 %	85 %
Room humidity ratio	2.2 g/kg	9.3 g/kg
Outdoor dry bulb	—	3.3 °C
Outdoor RH	—	100 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (sensible)	0	105
Equipment (sensible)	0	85
Infiltration / ventilation	-136	-74
Building heat transfer	-855	-571
External conduction — opaque	-99	-65

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	0	77

Zero at both peaks (not listed): Solar, Lighting, Air movement (inter-zone), External conduction — glazing, Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak sensible load	638 W	614 W
Peak hour	00:00–01:00	15 Aug 19:00–20:00 (HOY 5444)
Room dry bulb	20.2 °C	21.7 °C
Room resultant	21.2 °C	21.6 °C
Room RH	100 %	100 %
Room humidity ratio	15.0 g/kg	16.5 g/kg
Outdoor dry bulb	—	19.9 °C
Outdoor RH	—	79 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Lighting	0	60
Occupancy (sensible)	105	150
Equipment (sensible)	85	450
Infiltration / ventilation	-6	-11
Building heat transfer	415	92
External conduction — opaque	39	-126

LATENT COMPONENTS AT PEAK

Component	Design day (W)	Full year (W)
Occupancy (latent)	77	110

Zero at both peaks (not listed): Solar, Air movement (inter-zone), External conduction — glazing, Air handling unit, Equipment (latent)
Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	SAM.Analytical.Tas	SAM.Analytical.Tas
Read into model	27 Sep 2026 20:42	27 Sep 2026 20:42
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).
Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.